

Keep-beside-you reference card

Symbol / rule	Meaning and basis
x, z	Local and overall bulk B atom fractions, dimensionless
g_b, G	Bulk Gibbs energy J/mol atoms; total Gibbs energy J
R, T	8.3145 J/(mol K), 1000 K in this invented Day 2 model
$g'_b = dg_b/dx$	Slope of the bulk curve; equals $\mu_B - \mu_A$, the A-for-B exchange at fixed occupied sites, J/mol atoms
x_b, θ	Bulk B fraction and boundary B-site fraction; not area fractions
N_b, N_s, a_{gb}, n_{gb}	8000 bulk sites; 100 sites and 20 nm ² per boundary; $n_{gb} = 2$ boundaries
N_{Av}	$6.02214076 \times 10^{23} \text{ mol}^{-1}$; converts atom counts to mol
Γ_B	Chosen equal-site B excess, $5(\theta - x_b)$ atom/nm ² here
Open	Reservoir fixes x_b, μ_A, μ_B ; cell A/B may exchange
Closed	Cell fixes $B_{\text{tot}}, A_{\text{tot}}$; x_b changes with θ
$\bar{g}, D$	Closed energy J/mol all cell sites; $D = \bar{g}_{II}^* - \bar{g}_I^*$ at one inventory

Useful supplied arithmetic: $\ln(0.10/0.90) = -2.19722458$, $\ln(0.20/0.80) = -1.38629436$, $\exp(5000/8314.5) = 1.82459687$, and $1 \text{ nm}^2 = 10^{-18} \text{ m}^2$. No solver or live internet is needed. The source/contract lineage and optional detailed lessons are in the source map.